

光栅衍射

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1 实验数据

$leftY_2$	$leftY_1$	$leftG_1$	center	$rightG_1$	$rightY_1$	$rightY_2$
$255^{\circ}35'$	$255^{\circ}38'$	$256^{\circ}9'$	$265^{\circ}19'$	$274^{\circ}27'$	$274^{\circ}58'$	275°
$95^{\circ}31'$	$95^{\circ}28'$	$94^{\circ}57'$	$85^{\circ}47'$	$76^{\circ}39'$	$76^{\circ}08'$	$76^{\circ}06'$

2 数据处理

$$\theta_{green1} = 9^{\circ}9', \theta_{yellow1} = 9^{\circ}40', \theta_{yellow2} = 9^{\circ}42.5'$$

低压 Hg 灯的绿光波长为 $\lambda = 546.07nm$, 由

$$d = \frac{k\lambda_{green}}{\sin \theta_k} = \frac{1 \times 546.07nm}{\sin(9^{\circ}9')} = 3.4339773 \times 10^{-6}m$$

$$\nu = \frac{1}{d} = \frac{1}{3.4279752 \times 10^{-6}m} = 2.9120751 \times 10^5 m^{-1}$$

$$\lambda_{yellow1} = \frac{d \sin \theta_k}{k} = \frac{\sin \theta_{yellow1}}{\sin \theta_{green1}} \lambda_{green} = 576.61nm$$

$$\lambda_{yellow2} = \frac{d \sin \theta_k}{k} = \frac{\sin \theta_{yellow2}}{\sin \theta_{green2}} \lambda_{green} = 579.08nm$$

我们查阅标准值可知, $\lambda_{yellow1}^{标准} = 576.96nm$, $\lambda_{yellow2}^{标准} = 579.07nm$

$$\text{黄光 1 相对误差} = \left| \frac{\lambda_{yellow1}^{标准} - \lambda_{yellow1}}{\lambda_{yellow1}^{标准}} \right| \times 100\% = \left| \frac{576.96nm - 576.61nm}{576.96nm} \right| \times 100\% = 0.06\%$$

$$\text{黄光 2 相对误差} = \left| \frac{\lambda_{yellow2}^{标准} - \lambda_{yellow2}}{\lambda_{yellow2}^{标准}} \right| \times 100\% = \left| \frac{579.07nm - 579.08nm}{579.07nm} \right| \times 100\% = 0.00173\%$$

$$D_{\theta G1} = \frac{k}{d \cos \theta_k} = 294960.7699m^{-1}$$

$$D_{\theta Y1} = \frac{k}{d \cos \theta_k} = 295401.83m^{-1}$$

$$D_{\theta Y1} = \frac{k}{d \cos \theta_k} = 295438.5043m^{-1}$$

由于黄光可以辨别, 则

$$\Delta\lambda = \lambda_{yellow1} - \lambda_{yellow2} = 2.47nm$$

$$R \geq \frac{\bar{\lambda}_{yellow}}{\Delta\lambda} = 233.9453441$$